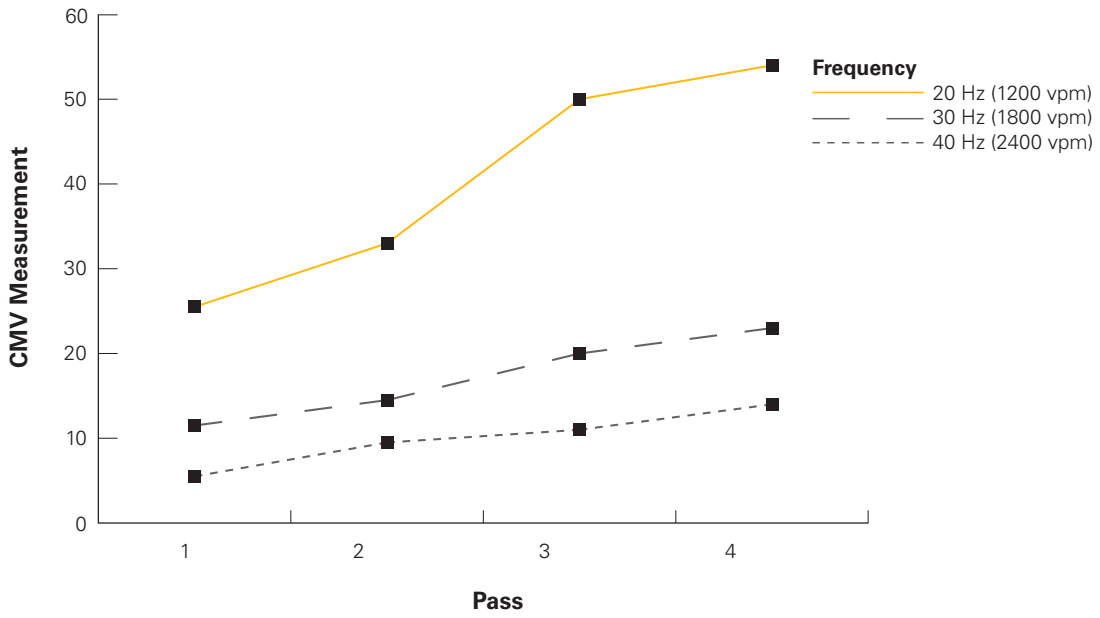


2. **Frequency –** Cat Compaction Control with CMV uses a ratio of the drum vibratory frequency and a measurement of the ground frequency response back into the drum at two times drum frequency to calculate the compaction value (CMV). Changing the vibratory frequency setting of the soil compactor will subsequently change the resulting measurement, even if the soil stiffness remains the same. This is because on a soil of a given stiffness, the CMV measurement will tend to read higher if measured at low frequency, and read lower if measured at high frequency. The reasons for this are complex, and related to the ratio of the soil's natural frequency to the frequency of the vibrating drum.

   The chart illustrates how changing the frequency can affect the CMV measurement. If all other machine and soil parameters are the same (e.g. ground speed, amplitude, soil type, etc.), there is a significant difference between the measured CMVs for each frequency setting. The frequency you use will affect the CMV measurement. This is true regardless of the stiffness of the soil.

3. **Ground Speed –** Machine-integrated compaction measurement results are affected by ground speed to some extent. Generally, slower speeds allow the vibrating drum to have more contact with soil, building soil stiffness sooner and at a deeper level, resulting in fewer passes and less overall time to meet compaction requirements. Data suggests that faster ground speed generally lowers the results for CMV, but can also increase the results for MDP. It is difficult to quantify precisely how much, because variations in soil type, water content and other factors cloud the issue during testing.

   When multiple passes are required to achieve final compaction levels (indicated by soil stiffness), the most efficient method is to use a slow ground speed and let the vibrator work the soil underneath the drum. The slower speed allows the vibrating drum to have closer impact spacing and more blows per distance traveled—and it will take fewer passes—resulting in less overall time to meet compaction requirements.

   There are also fuel efficiencies: two low-speed passes will consume less fuel than six high-speed passes. General rule of thumb: go slowly to compact quickly and maximize efficiency, but not so slow that de-coupling or de-compaction occurs. With machine-integrated compaction measurement technology, it is recommended to maintain a suitable consistent speed of about 1-2.5 km/h (0.62-1.5 mph) on rock and clay, and 2-5 km/h (1.2-3.1 mph) on sand and gravel. Automatic Speed Control on Cat B-Series compactors can be utilized to help with this factor.

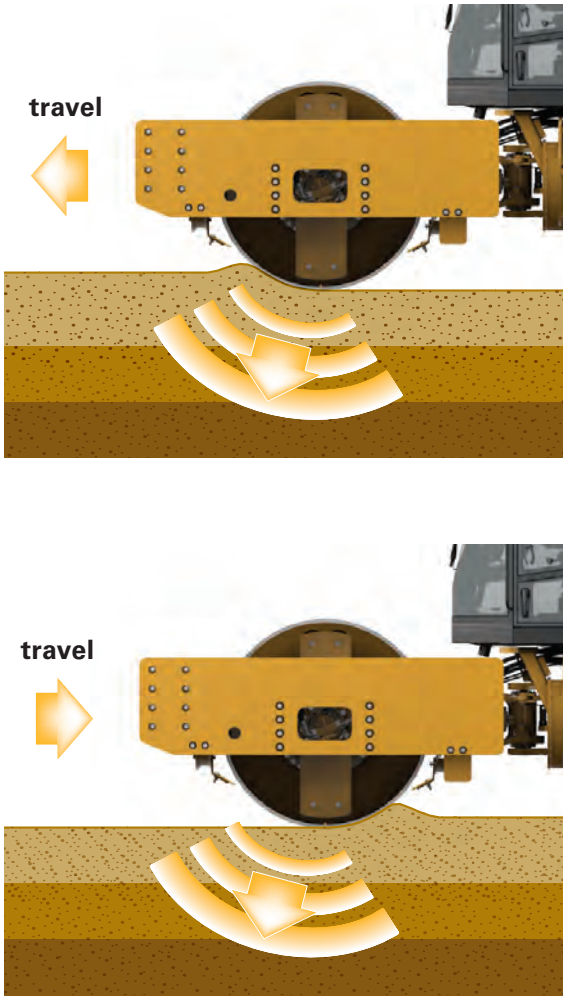
4. **Direction of Travel –** The direction of travel, forward or reverse, has an effect on the value of the machine-integrated compaction measurement data recorded for a given soil stiffness. The amount of variation between forward and reverse measurements typically ranges from 5 percent to 20 percent with higher variation on extremely soft soils.

   Machine-integrated compaction measurement results differ depending on travel direction because the rotation of the eccentric weight inside the drum either adds to or subtracts from the net torque applied to the drum as it is propelling over the soil (also due to balance of weight, wheels in front/behind the drum, and other factors). This torque influences the direction of the effective vibration into the ground and will cause the drum to sense either more toward the region already compacted or toward the softer area yet to be compacted.

5. **Vibratory State –** An active vibration system can also affect machine-integrated compaction measurement values. The impact may or may not be significant depending on the material. For test data, technologies that allow you to measure with the vibratory system off produce more reliable results because there are fewer variables influencing the measurement. Energy-based technologies like MDP allow you to do this.

6. **De-coupling or Double Jump –** When the stiffness of the soil increases, its natural or resonant frequency will come closer to matching the drum vibratory frequency. When this happens, the drum begins to completely bounce off the surface of the soil at half of the drum vibratory frequency, and the measured RMV (Resonance Meter Value) will increase. RMV is simply a measure of how much the drum is de-coupling. The more the machine is de-coupling, the less reliable the CMV measurement is.

DIRECTION OF TRAVEL



When the direction of the eccentric moment matches the direction of machine travel (top illustration), the net torque applied to the drum causes the accelerometer to sense the soil from a certain angle. When the direction of the eccentric moment does not match the direction of machine travel (bottom illustration), the torque forces change, changing the angle of measurement. This will result in consistently different CMV readings for forward and reverse operation.

[MACHINE-INTEGRATED COMPACTION MEASUREMENT PROCEDURES]

Soil compactors fitted with integrated compaction measurement technology have the capability to measure factors that can give operators a real-time, in-cab indication of soil stiffness. Many variables factor into the efficacy of this method and directly influence the consistency of the measurement. For that reason, the measurements are often verified utilizing one of the methods listed previously or other portable testing equipment. As the systems become more advanced, their use is better understood, and the results more accepted.

Machine-integrated compaction measurement typically has two modes of operation: production mode and proofing mode.

Production Mode – Production mode is utilized during the initial process of compaction. The intent of this mode of operation is to get as much soil compacted to an acceptable level as quickly and efficiently as possible. During production mode, machine-integrated compaction measurement technology provides the operator with a real-time indication of soil stiffness, as well as identifies areas of poor compaction where action may be necessary to bring the density up to specification.

In production mode, the machine amplitude is typically at high setting and the ground is only compacted until a nominal target value is reached. The operator monitors the display to see what areas have been adequately compacted. The primary concern is getting the job done efficiently, compacting as much fill as possible while not over-compacting areas that are sufficiently stiff.

Because the machine is operating in high amplitude, de-coupling may occur in spots. Due to all the variables—ground speed, direction, de-coupling and soil moisture variation—the integrated compaction measurement values seen during this mode of operation normally have more variation than what is possible to achieve. This may be considered the rough-cut approach to compaction measurement and the lack of accurate or less-variable data in these areas is not so important at this time.

Proofing Mode – Once production mode compaction has been completed, the compaction measurement system can be used as a proof roller to verify the quality of the work in proofing mode. Typically, this process is more precise than production mode. Many variables are controlled and kept consistent, including speed and direction of travel. This helps to ensure that these variables are not influencing the measurement.

The proofing mode is used when the contracting authority is requesting "in-field" data showing an accurate indication of ground stiffness for a particular phase or area of construction. This procedure may be done at periodic intervals of construction when it is convenient to run the compactor over a completed phase of the project in a controlled manner.

To achieve accurate results, the operator needs to keep all variables as constant as possible. This can be considered to be the precision phase of soil compaction.



RECOMMENDED PROCESS FOR PROOFING

1. Stake out the jobsite section to be tested and plan a rolling pattern that allows the operator to propel in the forward direction throughout the compaction testing operation.

2. Determine a target ground speed between 2.5-4 km/h (1.5-2.5 mph) that you will be able to maintain. Slower is better and use of automatic speed control will optimize uniformity of speed which allows for better data and compaction.

3. Set the amplitude to low. This will lessen the chance of drum de-coupling and will result in a measurement that does not penetrate as deeply into the soil. This makes correlation with other test methods easier.

4. Start vibration (or static rolling with MDP) and propel to begin compaction measurement at constant ground speed, amplitude, and frequency while driving forward.

5. Drum passes should be just touching or barely overlapped. Overlap at the ends or turnaround areas. *Note: overlapped areas can be considered as multiple passes and can cause discrepancies in the data.*

6. Manual collection of data can be cumbersome and needs to be matched up with position on the jobsite as accurately as possible within the staked out sections. Most machine-integrated compaction measurement systems do not allow for automatic data storage without the GNSS (GPS) option. Use spreadsheet software such as Excel to sort the data by direction traveled, using only the forward data for analysis. Physically mark out any locations deemed important enough to review or test with a correlation to a known field compaction measurement method.

7. For improved quality, measure the moisture content of the soil in a grid pattern over the entire area that was measured and compacted. Grid size can be adjusted to jobsite scale and requirements of the contracting agency. This will provide additional information for analysis of the compaction values, and the grid pattern will allow development of a soil moisture content iso-bar map. Soil samples for moisture measurements should be taken as soon as possible after the compactor has finished an area.

8. Review the manually recorded data and select those areas that you wish to correlate to another form of compaction measurement device. Select areas that read high values, low values and intermediate values. Select several areas of each.

9. Conduct correlation tests for previously marked locations. Don't approximate, as soil conditions can vary widely in just a short distance.

NOTE: *In order to fully leverage the capabilities of the machine-integrated technology, GNSS (GPS) mapping capabilities would be necessary to record all data with corresponding positioning detail on the jobsite. Refer to a later section of this guide for further details about "Intelligent Compaction."*

Summary of Machine-Integrated Compaction Measurement – On any compaction project, quality and cost are major concerns. Achieving applicable compaction targets efficiently and effectively is of the utmost importance. There are many specifications and historically based field compaction measurement methods. With machine-integrated compaction measurement methods, operators now have better tools that will ensure the best possible compaction quality at the lowest cost.

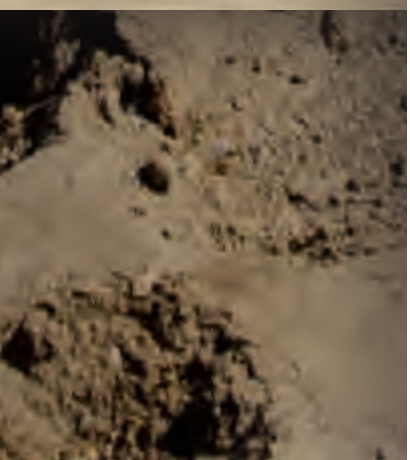
Machine-integrated compaction measurement is a wonderful technology when properly applied, but there are limitations to what it can do. Machine-integrated compaction measurement cannot tell you what soil type you are operating on, nor can it describe a soil's moisture content or physical characteristics.

Machine-integrated compaction measurement technology measures the responsiveness of the soil to provide a snapshot of the load-bearing capabilities. When properly set up and operated, a vibratory soil compactor with integrated compaction measurement provides information that the operator could not otherwise obtain. This information allows the trained operator to deduce the state of the soil. The data is an indication of the soil's stiffness, but not a guarantee. There are simply too many variables. However, a trained operator will understand what the measurements indicate and the course of action dictated by those measurements. The processes utilized are often more important than the technology itself.


CAT CS74B
CAT







Unit 5
INTELLIGENT COMPACTION

Intelligent compaction is the latest advancement in the application of vibratory soil compactors. The ability to accurately measure compaction, correlate measurements with GNSS coordinates, display them on a site map in the operator's compartment, and record and store data for documentation, was heretofore unimaginable. Only time will tell what technological innovations will come next. Caterpillar will be at the forefront of that discussion and the resulting discoveries.

[WHAT IS INTELLIGENT COMPACTION?]

Presently and in the recent past, intelligent compaction is a term applied to compaction measurement systems incorporated with a vibratory soil compactor. Definitions of intelligent compaction vary between different governmental agencies and equipment manufacturers. Generally speaking, intelligent compaction can be defined as a compactor-integrated technology applied to the compaction process that improves worksite efficiencies by eliminating human guesswork. The technologies provide operators with real-time compaction information that helps them determine when compaction is progressing and/or complete.

With this definition in mind, it is safe to say that the previously described machine-integrated compaction measurement technology is indeed a form of intelligent compaction. The integrated systems on soil compactors provide details about the compaction effort on a jobsite in real-time, enabling operators and jobsite superintendents to access information that they did not previously have.

More sophisticated systems can also map out the data to provide a visual perspective of the work completed, as well as store data for later analysis.

One example of an early specification for a State Department of Transportation (USA) is:

> **Intelligent Compaction (IC)**
> This process involves measuring **and recording the time, location and compaction parameters** of the granular treatment during the compaction process with a vibratory roller that is equipped with an accelerometer-based measuring system and global positioning system.

They now have a separate definition or requirement for what an intelligent compactor is:

> **Intelligent Compaction (IC) Roller**
> Rollers shall be vibratory rollers with an accelerometer-based measurement system and **capable of recording the compaction parameter measurements**.

At the same time, the U.S. Federal Highway Administration (FHWA) describes intelligent compaction as the following:

> **Intelligent Compaction (IC)** refers to the compaction of road materials, such as soils, aggregate bases, or asphalt pavement materials, using modern vibratory rollers equipped with an integrated measurement system, **an onboard computer reporting system, Global Positioning System (GPS) based mapping**, and optional feedback control. IC rollers facilitate real-time compaction monitoring and timely adjustments to the compaction process by integrating measurement, **documentation**, and control systems. IC rollers also maintain a continuous record of color-coded plots, allowing the user to view plots of the precise location of the roller, the number of roller passes, and material stiffness measurements.

The European Union has also created a means to define the use of intelligent compaction. In their pamphlet "Guidelines to evaluate soil and asphalt compactors equipped with continuous compaction control (CCC)," the Committee for European Construction Equipment (CECE) has created a matrix to classify equipment with continuous compaction control technologies. (See Appendix.)



It is important to note that more recent definitions of intelligent compaction are clearly defining not only the capabilities of integrated compaction measurement and real-time display to the operator, but also the capability of recording positioning data and storing it for documentation, further analysis and records retention. As such, the definition of intelligent compaction is continually evolving.

Caterpillar believes an intelligent compactor should measure compaction, correlate measurements with GNSS coordinates, display a map of the measurements, record the data and document the results. These capabilities provide many benefits in terms of time and cost for the operators, contractors, and project owners. Therefore, the current definition used by Caterpillar for intelligent compaction is as follows:



Intelligent Compaction (IC)
A system that measures soil compaction, displays the measurements to the operator, records and maps the compaction results using a GNSS mapping system, and controls or guides the machine compaction effort in response to the measurement system.

This definition can be used for both vibratory and non-vibratory compactors and does not require an accelerometer-based measurement system. As described previously, MDP is a new technology and has many advantages over accelerometer-based compaction measurement technology depending on application.

Ongoing, this guide will only refer to intelligent compaction in cases where the compactor is equipped with integrated compaction measurement capabilities (either CMV or MDP), mapping capabilities and the ability to record and store data for documentation and later analysis off-board the machine.

Positioning of the Compactor on the Jobsite

Machine-integrated compaction measurement technology can be augmented with Global Navigation Satellite Systems (GNSS) technology, which enables accurate jobsite positioning through the use of different constellations of satellites in the sky. GNSS technology is widely available and offers different levels of accuracy, some requiring off-board infrastructure providing positioning correction data.

With this level of data and detail, the compaction measurement technology can now be matched up with physical location on a jobsite and can be configured to map these values, including pass counts, direction of machine travel and many other machine settings.

Regardless of the machine-integrated compaction measurement technology employed, the system provides real-time measurement of the soil that is being compacted at any given time. Adding mapping functionality, the ability to record and plot measurements to their precise location on a map makes the information much more useful.

Positioning Data, how does it work? Mapping systems utilize the Global Navigation Satellite System (GNSS) to provide mapping data for each measurement recorded. This includes GPS (operated by the U.S. Department of Defense) and GLONASS (operated by the Russian government), as well as other systems due to come online in the future, including Galileo in the EU and Compass in China.

Location is plotted by triangulation with the known position of satellites from these systems. The satellite systems are not accurate enough to be practical without some degree of correction. The mapping systems on soil compactors use augmentation to correct the satellite signals and provide a useful degree of accuracy. There are two predominant forms of augmentation available: SBAS and RTK.

Most systems utilize SBAS, or Satellite Based Augmentation System, to correct satellite positioning signals. SBAS triangulates to multiple ground-based sites that provide known "anchor points" from which to provide a correction measurement. SBAS systems are typically accurate up to 1 m (3 ft) and require no off-board infrastructure.

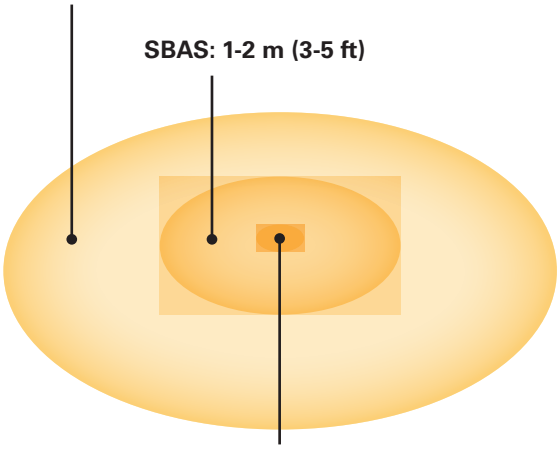
Alternatively, many manufacturers can augment the signal with RTK, or Real Time Kinematic, correction. This technology requires the use of local radio base stations to provide correction data.

In fact, recent technology also allows for RTK accuracies via cell or modem technologies or even Virtual Reference Stations (VRS)—but this requires a more IT savvy organization and support. The base stations are expensive, and the technology requires line-of-site from the receiver on the compactor to the base station or mobile units. However, RTK provides more accuracy than SBAS, down to a few centimeters (inches). It also allows the system to record elevation data, which would enable the compactor to map grade elevations as well.

This provides a substantial benefit, as the compactor is often the last machine to operate on an earthworks jobsite and can save significant cost/time in final elevation surveying activities.

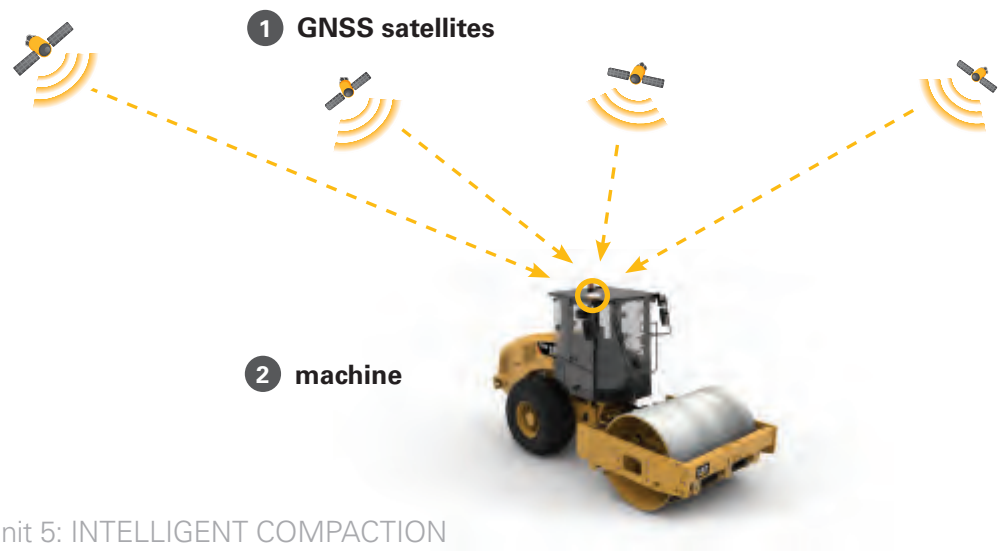
SATELLITE SYSTEM ACCURACY

Autonomous: 10 m (30 ft)
no correction

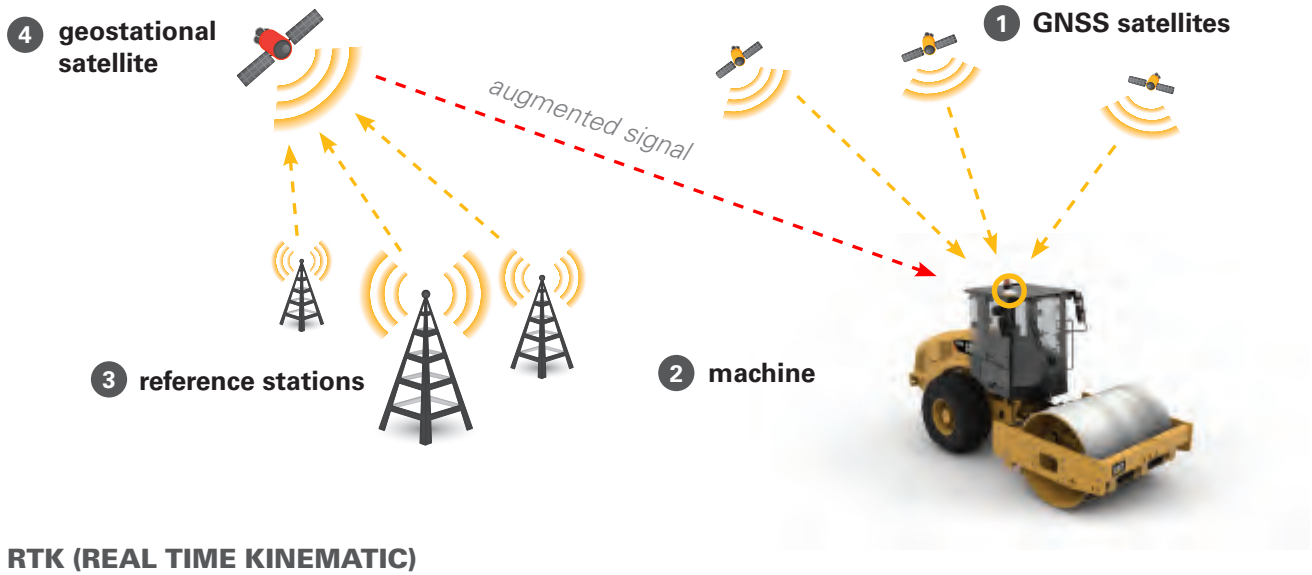


RTK: 1 cm (0.4 in) horizontal, 2 cm (0.8 in) vertical
(with local base or with VRS)

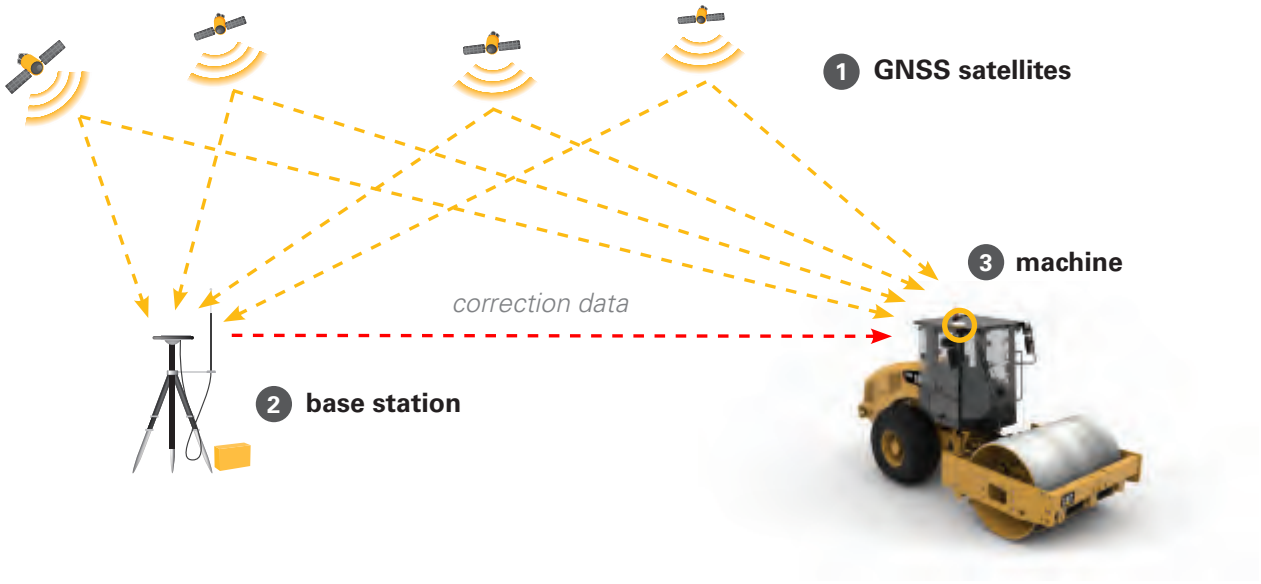
AUTONOMOUS



SBAS (SATELLITE - BASED AUGMENTATION SYSTEM)



RTK (REAL TIME KINEMATIC)



Benefits of Positioning Data

By itself, machine-integrated compaction measurement can reveal many things about the state of compaction in real-time; but this information is very specific, reflecting a momentary snapshot in time. Positioning data allows the system to provide not just a single measurement as it is taken, but ALL measurements in the context of where they were taken. This changes the view from momentary to encompassing, and opens the door for in-depth analysis. Suddenly, the operator—as well as site management—can access a picture of compaction quality for the whole site, rather than just a snapshot in time.

This capability is a big differentiator for IC systems when compared with the other testing methods. Traditionally, ground personnel conduct quality testing using portable testing procedures in a few select locations. The process is time consuming and expensive. The results from the tests are used to represent a much larger area than what is actually tested, often at a ratio of 1:1 million—not exactly statistically reassuring. IC can measure the entire site in the time that it takes to roll over it.

Additionally, some systems are capable of importing engineering or architectural 3D designs into the on-board display. This can be advantageous on a jobsite where stakeless grade control is already in use or where grade stakes and other landmarks are not available.

Using the Collected Data

It is easy to see the advantages that the additional data offers the operator, enabling a more efficient and cost effective way to achieve quality compaction on a jobsite. Many inspectors and road administrations require some real-time and basic in-field reporting, which is text data documenting basic compaction progress via an in-cab printing application. But many authorities are focusing now on more detailed off-board reports. This requires the transmission of all recorded compaction data from the jobsite machine to a PC in an office. Data can be moved manually via jump drive or wirelessly via communications hardware and software.

Once all the data has been received on the office computer, users have the challenge of filtering and sorting the data to construct reporting documentation required by inspection and road authorities. There are many such software programs available today to assist in this task, including AccuGrade Office, SiteVision Office, VisionLink, Veda and many more. These software solutions range in file format types, capabilities and prices.

VISIONLINK INTERFACE



[BENEFITS OF INTELLIGENT COMPACTION]

Quality Control & Quality Assurance Documentation – Intelligent compaction allows for documentation of work completed. It also allows in-process control with the ability to monitor progress daily or near real-time, as well as electronic storage and analysis of results that can be correlated with long-term or historical records of the jobsite data.

Increased Operator Productivity – Output is visually displayed for the operator, helping to determine if the soil has reached target stiffness. This empowers the operator with real-time data that can be acted upon. For example, the system can alert the operator to soft spots, indicate potential moisture content issues, and, with RTK accuracy, the compactor can also check final jobsite grades and elevations.

More Efficient Jobsites – Output of results delivers a map of the entire compacted area, which can reveal areas that need more compaction or areas that are already completed. This minimizes passes and fuel consumption, and can quickly reduce the number of manual compaction tests needed on the jobsite, keeping production at a constant level while reducing testing costs with fewer samples to tag and store.

Confidence in Results – Accurate positioning identifies exact locations of compaction problems early in the construction process, allowing for more cost effective corrections and reduced risk of later rework. The data provides a reliable reference of the overall quality of the job in a visual, easy-to-understand format. This allows trained operators to deduce when a job is complete, giving them the confidence to move on to the next area rather than waiting for the results of conventional testing.

MEASURING RESULTS



[CURRENT SPECIFICATIONS FOR INTELLIGENT COMPACTION]

Intelligent compaction for soil applications has generally increased in use as governmental agencies have studied and accepted the benefits of the technology and produced specifications for its use. Specifications help to ensure that the technology is applied in a manner that produces an acceptable result for the specifying agency.

The U.S. Federal Highway Administration (FHWA) has developed a set of generic specifications for the utilization of intelligent compaction on soils. These specifications are intended for state departments of transportation to use as-is or modify to meet their requirements. Here is an example of the specifications as issued:

The IC rollers shall meet the following specific requirements:

1. IC rollers shall be self-propelled single-drum vibratory rollers equipped with accelerometers mounted in or about the drum to measure the interactions between the rollers and compacted materials in order to evaluate the applied compaction effort. IC rollers may be smooth or pad footed drums.

2. The output from the roller is designated as the Intelligent Compaction Measurement Value (IC-MV), which represents the stiffness of the materials based on the vibration of the roller drums and the resulting response from the underlying materials.

3. GPS radio and receiver units shall be mounted on each IC roller to monitor the drum locations and track the number of passes of the rollers.

4. The IC rollers shall include an integrated on-board documentation system that is capable of displaying real-time, color-coded maps of IC measurement values including the stiffness response values, location of the roller, number of roller passes, roller speeds, together with the vibration frequency and amplitude of roller drums.

5. The display unit shall be capable of transferring the data by means of a USB port.

6. An on-board printer capable of printing the identity of the roller, the date of measurements, construction area being mapped, percentage of the construction area mapped, target IC-MV, and areas not meeting the IC-MV target values. (Printer option to be selected by the each state DOT.)

Governmental authorities from other countries have developed their own specifications built around the road building process in their respective countries. While they may vary from the FHWA specifications, the goal is similar in that they want to provide a standard for use of the equipment.

Spec.	Equipment	Field Size	Location Specs	Documentation
Mn/DOT (USA)	Smooth drum or padfoot vibratory roller (25,000 lbs.)	100 m x 10 m (minimum at base). Max 1.2 m thick.	One calibration/ control strip per type or source of grading material	Compaction, stiffness, moisture, QC activities, and corrective actions (weekly report)
ISSMGE (International Society of Soil Mechanics and Geotechnical Engineering) **www.issmge.org/**	Roller chosen by experience	100 m by the width of the site	Homogenous, even surface. Track overlap ≤ 10% drum width.	Rolling pattern, sequence of compaction and measuring passes; amplitude, speed, dynamic measuring values, frequency, jump operation, and corresponding locations
Earthworks (Austria)	Vibrating roller compactors with rubber wheels and smooth drums suggested	100 m long by the width of the site	No inhomogeneities close to surface (materials or water content). Track overlap ≤ 10% drum width.	Compaction run plan, sequence of compaction and measurement runs, velocity, amplitude, frequency, speed, dynamic measuring values, jump operation, and corresponding locations
Research Society for Roadand Traffic (Germany)	Self-propelled rollers with rubber tire drive are preferred; towed vibratory rollers with towing vehicle are suitable.	Each calibration area must cover at least 3 partial fields ~ 20 m. long	Level and free of puddles. Similar soil type, water content, layer thickness, and bearing capacity of support layers. Track overlap ≤ 10% machine width.	Dynamic measuring value; frequency; speed; jump operation; amplitude; distance; time of measurement; roller type; soil type; water content; layer thickness; date, time, file name, or registration number; weather conditions; position of test tracks and rolling direction; absolute height or application position; local conditions and embankments in marginal areas; machine parameters; and perceived deviations
Vägverket (Sweden)	Vibratory or oscillating single-drum roller. Min. linear load 15–30 kN.	Thickness of largest layer 0.2–0.6 m.	Layer shall be homogenous and non-frozen. Protective layers < 0.5 m may be compacted with sub-base.	—

Compaction Specs	Speed	Frequency
90% of the roller compaction measurements and average of LWD modulus measurements (based on 3 tests) must be at 90% of the target values established in the calibration strip.	Same during calibration and production compaction	
Correlation coefficient ≥ 0.7. Minimum value ≥ 95% of Ev1, and mean should be ≥ 105% (or ≥ 100% during jump mode). Dynamic measuring values should be lower than the specified minimum for ≤ 10% of the track. Measured minimum should be ≥ 80% of the specified minimum. Standard deviation (of the mean) must be ≤ 20% in one pass.	Constant 2–6 km/h (± 0.2 km/h)	Constant (± 2 Hz)
Correlation coefficient ≥ 0.7. Minimum value ≥ 95% of Ev1, and median should be ≥ 105% (or ≥ 100% during jump mode). Dynamic measuring values should be lower than the specified minimum for ≤ 10% of the track. Measured minimum should be ≥ 80% of the set minimum. Measured maximum in a run cannot exceed the set maximum (150% of the determined minimum).Standard deviation (of the median) must be ≤ 20% in one pass.	Constant 2–6 km/h (± 0.2 km/h)	Constant (± 2 Hz)
The correlation coefficient resulting from a regression analysis must be ≥ 0.7. Individual area units (the width of the roller drum) must have a dynamic measuring value within 10% of adjacent area to be suitable for calibration.	Constant	
Bearing capacity or degree of compaction requirements may be met. Mean of compaction values for two inspection points ≥ 89% for sub-base under roadbase and for protective layers over 0.5 m thick; mean should be ≥ 90% for roadbases. Required mean for two bearing capacity ratios varies depending on layer type.		Constant 2.5–4.0 km/h

[INTELLIGENT COMPACTION OPERATION PROCEDURES]

Similar to the use of only the machine-integrated compaction measurement technology discussed previously, intelligent compaction-enabled vibratory soil compactors have similar processes to follow, with slight differences due to the additional capabilities the more advanced systems provide. There are two modes of operation: production mode and proofing mode.

Production Mode – Again, the intent of this mode of operation is to compact as much soil to an acceptable level as quickly and efficiently as possible in both forward and reverse directions until a nominal target value is reached. The compaction is conducted in normal working parameters—travel speed of 1 to 2,5 km/h (0.6-1.6 mph) on rock fill and clay; 2-5 km/h (1.2-3.1 mph) on non-cohesive sands and silt soils—and the machine amplitude is typically at high setting.

In the case of an intelligent compaction equipped compactor, the operator can now receive input other than a simple dimensionless number giving an indication of soil stiffness or load bearing strength. With the additional color display and data collecting and storage capabilities, the operator has a snapshot of compaction in the exact location of the compactor and can monitor the jobsite by pass count, percent of a target compaction measurement value, or even by percent of change of the compaction target value on a pass-by-pass basis (for both areas of sufficient as well as areas of poor compaction). As necessary, the operator can also digitally mark points and areas where action may be necessary to bring the compaction up to specification. All stored data can be taken off-board to review, filter and analyze as necessary.

In this mode of operation, the mapping system provides the operator with a visualization of jobsite compaction progress and a moderate level of quality assurance. However, because of all the variables—ground speed, direction, de-coupling, soil moisture variation—the compaction data taken during this mode of operation is still of lower quality than what is possible to achieve. However, it still gives the operator data to maximize efficiency and uniformity and eliminate the reliance on guesswork during the compaction process.

When the values reach a range near the target value, the operator can move on to a new area while quality personnel conduct testing to ensure the work meets specification or the area can be made ready for a proofing mode pass.



Proofing Mode – Once production compaction has been completed, the compaction measurement system can be used as a proof compactor to verify the quality of the work in this more precise mode of operation. This approach is intended for use when the contracting authority is requesting documentation or a map showing an accurate indication of ground stiffness for a particular phase of construction.

It is of utmost importance to ensure that variables are controlled and kept consistent, thus working parameters are now constant travel speed of 3 km/h (2 mph) with machine amplitude in low setting (or off if necessary in the case of over compaction or de-coupling or other issues with an MDP equipped machine). This helps to ensure that these variables are not influencing the measurement and data being collected and stored for reporting purposes.

It should also be noted that when it comes to proof rolling, what you are measuring—the degree of compaction of the soil—is changing as you measure it. As the heavy compactor moves, its static weight is introducing pressure and other forces into the soil below. For this reason, operators need to take care to diminish the amount of force introduced as they are measuring. The ability of energy-based systems to measure with a static drum (without vibration) makes them ideal for this application.



RECOMMENDED PROCESS FOR PROOFING

1. Plan a rolling pattern that allows the operator to drive in the forward direction throughout the mapping operation.

2. Determine a target ground speed between 2.5 to 4 km/h (1.5 to 2.5 mph) that you will be able to maintain. Slower is better, and use of automatic speed control will allow uniformity of speed, which enables better data measurement and compaction.

3. Set the amplitude to low. This will lessen the chance of drum de-coupling and will result in a measurement that does not penetrate as deeply into the soil. This makes correlation with other test methods more likely.

4. Select the "proofing on" from the display menu and start vibration (or static rolling with MDP) and propel to begin compaction measurement at constant ground speed, amplitude, and frequency while driving forward.

5. Use the GNSS positioning capabilities to guide the machine so that only one pass is made over the area of interest. Drum passes should be just touching or barely overlapped. Overlap at the ends or turnaround areas. NOTE: overlapped areas can be considered as multiple passes and can cause discrepancies in the recorded data.

6. When the entire area has been covered, select "proofing off" on the display.

7. For improved quality, measure the moisture content of the soil in a grid pattern over the entire area that was measured and compacted. Grid size can be adjusted to jobsite scale and requirements of the contracting agency. This will provide additional information for analysis of the compaction values, and the grid pattern will allow development of a soil moisture content iso-bar map. Soil samples for moisture measurements should be taken as soon as possible after the compactor has finished an area.

8. Review the compaction map and data to select those areas that you wish to correlate to another form of compaction measurement device. Select areas that read high values, low values and intermediate values. Select several areas of each.

9. Conduct correlation tests using a hand held GNSS rover to locate the selected test areas as accurately as possible. Don't approximate, as soil conditions can vary widely in just a short distance. (See discussion of correlating test equipment in the next section.)

[ECHELON PATTERN WITH TWO COMPACTORS]

Several methods can be used to calibrate a target value for compaction measurement (CMV or MDP) and the required number of passes. Some of these methods will be dictated by the contracting agency, and may differ from what is printed in this guide. The industry is moving toward a practice of emphasizing that the area be uniformly compacted to an acceptable level, and is moving away from trying to obtain a specific compaction value or density for all areas of the worksite.

Site Calibration Using a Test Strip and an Independent Testing Device

This method is intended to eliminate as many variables in the measurement process as possible, and to use the same soils and methods as are going to be used during the actual construction of the road or building site. It is very time and resource consuming, but is the best way to understand the process and build a baseline understanding of the technologies involved.

1. Locate a test area that can be left intact for the duration of the project and has similar soils, grades and subsoil structure as the majority of the project.

2. Excavate/fill the test area to grade and level to create the sub-base for the test strip.

3. Bring in the vibratory soil compactor and create a base line compaction map using the intelligent compaction system in proofing mode and low amplitude, constant frequency (or with vibration off in the case of MDP), constant forward direction at 3 km/h (2 mph).

4. If the compaction map shows high variability (90 percent of the values should be within 20 percent variation of the average value), try compacting the base in high amplitude to bring the compaction level of the soft areas up to be more equal to the higher value areas. Repeat step #3.

5. If the sub-base is now uniformly compacted (meeting the test: 90 percent of the values within 20 percent variation of the average value), go on to step #6. If not, either a new test strip site needs to be selected (step #1) or remedial work needs to be done to the sub-base to obtain a more uniform map. Remedial work may involve excavating hidden objects such as rocks and clay balls or stabilizing the soil with lime or some other agent. If a stabilizing agent is used on the test strip, it must also be used throughout the jobsite for the test strip to have validity.

6. Measure the compaction of the sub-base with a portable plate load test or light falling weight deflectometer using a uniform pattern of test points across the full area of the test strip. Avoid using nuclear density or sand cone density devices, as they measure a different soil characteristic (density) than the vibratory compactor (stiffness). These test points need to be located using a precision GNSS device capable of decimeter accuracy in order to be correlated correctly to the GNSS compaction map obtained from the intelligent compactor. As an alternative to testing in a grid pattern, use the compactor GNSS map and pick several scattered locations that represent the high, medium and low compaction values. Obtain at least three tests for each range of values (9 test points minimum). Using more test points provides a more statistically accurate result.

7. Measure the moisture content at each of the test points. If the moisture content shows wide variation, the correlations will not produce uniform results.

As a cautionary note, remember that the portable test device used to calibrate the compactor-integrated compaction measurement value has its own variability of measurement. That means that it could be used to measure the same soil with the same properties several times and not provide the exact same results. For example, nuclear density measurements can vary by 15 percent on the same sample. Current practice is to take a reading, rotate the device 90 degrees and take another reading. The average of the two readings is recorded as the measured value.

8. Correlate compaction test values to those obtained from the GNSS map on the compactor and graph the results with compaction measurement values on the y-axis and portable field test results on the x-axis. This is called a scatter plot.

9. Use curve-fit methodology to determine the best calibration curve between the integrated compaction measurement values and the field test method. This is the tool that will only be used for that soil type on the jobsite and tests of the sub-base compaction.

10. Bring the first lift of fill material and place uniformly at the specified depth.

11. Compact the material until it is uniformly compacted and the integrated compaction measurement values do not change much when traveling in the forward direction.

12. Repeat steps #2 through #9 for this lift of material. This base lift will have its own calibration curve that will be used throughout the jobsite for this lift of material.

13. Repeat steps #10 through #12 for all layers of fill that are brought in, each layer having its own calibration curve.

14. If the ground conditions change due to the weather, rerun a proof run on the test strip to re-establish what constitutes an acceptance level for the integrated compaction measurement value.



Calibration of Integrated Compaction Measurement Results and Pass Count Using a Soil Compactor (No Independent Field Test Device)

This process, or a similar one, is used in some Scandinavian countries. It is more pragmatic and demands less time and resources while providing a means of having good compaction control without using some other compaction measurement as a standard. The objective is to use the compactor to establish a nominal maximum level of compaction for the worksite materials, and determine the number of passes to reach that level. The purpose is to create a uniform level of compaction across the entire jobsite. The following process uses a test strip to set compaction baselines. A test strip is optional, as a portion of the worksite or the entire worksite can be compacted as if it were a test strip.

1. Locate a test area that can be left intact for the duration of the project that has similar soils, grades and subsoil structure as the majority of the project.

2. Excavate/fill the test area to grade and level to create the sub-base for the test strip.

3. Bring in the vibratory soil compactor and create a base line compaction map (proof map) using the intelligent compaction system in proofing mode and low amplitude, constant frequency (or with vibration off in the case of MDP), constant forward direction at 3 km/h (2 mph).

4. Check moisture content of the soil at multiple locations on the test strip. If it is too high or too low, correct before doing any additional compaction.

5. If the compaction map shows high variability (90 percent of the values should be within 20 percent variation of the average value), compact the base in high amplitude until the compaction measurement values are more consistent across the test strip. Repeat step #3.

6. Upon obtaining uniform compaction, check the moisture content at several locations and record the results.

7. Bring the first lift of fill material. Select the proof map feature and compact the fill layer using a high amplitude setting and slow uniform ground speed of 3 km/h (2 mph). Complete an up and back pass in each track before moving over to run on uncompacted material. Moisture content should be uniform and at an optimum level for the soil or rock fill being used.

8. Repeat the compaction cycle over the entire area again, noting the level of compaction measurement value that is most predominant for each compaction pass.



9. Keep repeating step #8 until the compaction level between passes does not change substantially, or the compactor starts to de-couple.

10. The average value that the compaction measurement begins to level off is the target value and the number of passes it took to reach this level of compaction is your target pass count.

11. Record these results, and set the display target value and pass count accordingly for this lift.

12. Repeat steps #7 through #11 with a new proof map for each new lift. When complete, you will have a target compaction measurement value and pass count for each layer.

13. If several of the layers possess nearly identical values, use a single target compaction measurement value for all corresponding layers of fill.

14. Compact the worksite as normally done, using the compaction map as a guide to achieve uniform compaction over the entire area.

15. With time and experience in a region, skilled operators may be able to set the target compaction measurement values and pass counts without utilizing a test strip.

16. If a more precise measure of the final compaction level is desired, use the compactor in the proofing mode (low amplitude, constant speed of 3 km/h [2 mph], constant frequency and forward direction only).





Use of Intelligent Compaction Values Without Calibration on the Jobsite

This process is the most pragmatic and demands little or no additional time. It must be noted that this process is best suited for what is described previously as a production mode of compaction and requires some experience with the system, an understanding of how the technology works and the process of soil compaction in general.

The objective of this process is to use the intelligent compaction technology to compare the relative change in compaction progress on a pass-by-pass basis in order to know when the physics behind the compactive effort is adequate for the conditions at hand. As stated earlier, using intelligent compaction and integrated compaction measurement technology is not a guarantee of compaction or density; and, often times, the process employed is more important than the tools and technology used on a jobsite. A compactor of given characteristics can become ineffectual after a certain number of passes on a certain material, rendering the compaction goal unachievable and making all subsequent passes wasteful. It would be beneficial to know when this condition takes place in order to stop wasting time and consuming fuel unnecessarily.

1. Size the vibratory soil compactor with intelligent compaction capabilities as best you can considering compaction targets, soil types, moisture levels, lift thickness, etc. Details provided earlier describe some factors to take into consideration when sizing and choosing the configuration of the compactor.

2. Begin compaction on the jobsite with the intelligent compaction display set up to map compaction measurement values and comparing the percent change from one pass to the next.

3. The mapping feature can be set up and customized to assign a certain color for a certain range of percent change, one pass to the next. As an example, set the display to map the color red for areas showing 50-100 percent change pass-to-pass, yellow for areas showing 10-49 percent change pass to pass, and finally green for 0-9 percent change pass to pass. These ranges can be changed according to experience or jobsite conditions as necessary.

4. Continue rolling and compacting with the focus on turning the map green.

5. If there are areas that cannot be brought to green (little to no change from one pass to the next), this is an indication that there may be some issue with the suitability of the soil in that area, or perhaps some sub-base concerns that need to be addressed.

6. When the map is sufficiently green and no further change in compaction measurement values are noted from one pass to the next, utilize a portable plate load test or light falling weight deflectometer as described in the examples. Be sure to use a uniform pattern of test points across the full area being compacted/tested or as desired to prove that the compaction will pass or fail the compaction target specified for the project.

7. If the tests show a passing level of compaction, continue with the process on the jobsite as described.

8. If the tests show that compaction targets are not met, one of two things is happening. 1) The machine size and operating weight are not correct for the soil type and lift thickness and/or 2) the moisture content of the soil is incorrect (either too dry or too wet). Either way, the compactor on the jobsite will not be capable of further compaction progress until some conditions are changed.

